

Muhammad Umar Khan

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Pinterest

Dear Hiring Manager,

I am writing to express my enthusiastic interest in the Machine Learning Intern position at Pinterest. As a Master's student in Computer Science with a strong background in AI/ML and a passion for creating impactful, data-driven solutions, I am excited by Pinterest's mission to inspire millions and the opportunity to contribute to your large-scale recommendation systems and visually rich experiences.

My project, MindForge AI Suite, demonstrates my ability to lead a project from start to finish, implementing a RAG pipeline with TF-IDF chunk retrieval and optimizing multi-document analysis throughput by ~75% using parallel summarization. Furthermore, in my Market Predictor project, I developed a custom CNN+LSTM neural network from scratch for stock and crypto price forecasting, integrating Claude AI for chart image analysis and investment insights, showcasing my proficiency in applying advanced ML techniques to solve complex problems.

My experience as a Research Assistant at the University of Windsor involved engineering a Python-based image segmentation pipeline and evaluating multiple CNN and Graph-CNN models for pixel classification, directly aligning with the need for cutting-edge research in machine learning. I am proficient in Python, TensorFlow, and PyTorch, and have experience with data analysis libraries like NumPy and Pandas, which would be valuable in collecting, analyzing, and synthesizing findings from data.

I am eager to leverage my skills in machine learning, natural language processing, and data analysis to contribute to Pinterest's innovative projects and help build personalized experiences for your users. I am confident that my proactive approach to problem-solving and my strong interest in applying AI to drive meaningful product innovation make me a strong candidate for this role. I would welcome the opportunity to discuss how my qualifications align with your team's needs.

Sincerely,

Muhammad Umar Khan